

Countdown 31 — Admin Panel & Operations Master Specification

Exhaustive, tab-by-tab engineering and architectural contract covering every single screen, form field, modal dialog, state transition, security policy, REST endpoint, and database operation in the Admin Panel.

Target Stack: NestJS 11 · Prisma 6 · PostgreSQL 16 · Redis 7 · Next.js 16 · **Lead / Author:** Muhammad Sameer Ali
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ADMIN PANEL ARCHITECTURAL MANDATE:

This document is dedicated exclusively to the Admin Console (/admin). It provides the definitive technical specification for all 6 primary administrative domains: **Dashboard Overview, Tournaments & Templates Manager, Sponsor Campaign Studio, Sponsor Accounts & Provisioning, Game Studio & Rules Engine, User Moderation & Ledger Auditing, and Global Platform Settings.**

1. Global Admin Architecture, Authentication & Navigation Topology

The Admin Console is strictly segregated from public game surfaces via cryptographic authentication guards and role boundaries:

Admin Tab / Route	Key Modules & Screens Hosted	Security & RBAC Enforcement
1. Overview /admin	Real-time operational KPIs (active users, player growth), verification pipeline for pending sponsor tournaments, and partner inquiry triage.	JWT + Role.ADMIN AdminGuard with active tokenVersion check.
2. Tournaments /admin/tournament	Tournaments roster table, create tournament modal, gate codes generator, campaign & cause templates gallery, 1-click tournament deployer, roster inspector.	JWT + Role.ADMIN Cascading cascade deletion permissions.
3. Campaign Studio /admin/tournament/{id}/campaign	Co-branding visual designer: Identity copy, visual theme, logo & 3D animations, charity cause goal tracking, multi-lane approval, media uploads, rollback.	AdminOrSponsorGuard Sponsors restricted strictly to owned tournament.
4. Sponsor Accounts /admin/sponsor	Sponsor provisioning console, automated credential generation, secret-box AES-256-GCM password encryption, credential copy & status controls.	JWT + Role.ADMIN Admin-issued credential escrow.
5. Game Studio /admin/game	7-tab game engine configurator: Control room preview, branding copy, arena world designer, gameplay timers, 2-skill deck editor, bot roster, menu builder.	JWT + Role.ADMIN Atomic JSON schema validation.
6. Users & Ledger /admin/users	User roster search/filter, ban/unban moderation, wallet balance adjustments (ADMIN_AUDIT), and immutable audit logs.	JWT + Role.ADMIN Double-entry ledger reconciliation.
7. Settings /admin/settings	Master admin profile identity, theme family switcher (Barnaby Pasture vs Moonlit Bazaar), maintenance mode, treasury wallet parameters.	JWT + Role.ADMIN Platform-wide broadcast trigger.

2. Tab 1: Operational Dashboard & Verification Pipeline (/admin)

The primary entry point aggregates live operational statistics and queues requiring immediate administrator action:

- **KPI Metrics Cards:**
 1. **Total Players:** Cumulative registered accounts in PostgreSQL (User.count()).
 2. **Active (7 Days):** Distinct users with match activity in the trailing 7 days.
 3. **New Today:** Accounts created since 00:00:00 UTC today.
 4. **Active Sponsors:** Total registered sponsor accounts in active status.
 5. **Pending Approvals:** Count of tournaments in status=PENDING awaiting review.
- **Pending Sponsor Tournaments Queue:** Table listing sponsor-submitted tournaments. Action buttons: **[Approve]** (POST /admin/promo-tournaments/{id}/status { action: 'approve' }) & **[Reject]** (POST ... { action: 'reject' }).
- **Partner Inquiry Pipeline:** Triage inbound sponsorship leads from the public landing page (Inquiry table). Status workflow: NEW → CONTACTED → CONTRACT_SIGNED → CLOSED. Admin can assign an inquiry to a newly created sponsor account in 1 click (POST /admin/inquiries/{id}/assign-sponsor).

3. Tab 2: Tournaments Manager, Gate Codes & Template Gallery (/admin/tournament)

The central hub for creating, filtering, deploying, and inspecting all platform tournaments:

1. Tournament Roster Table & Multi-Status Filters:

- **Filters:** `ALL`, `PENDING` (needs approval), `LIVE` (currently open/playing), `COMPLETED` (finished with winner).
- **Columns:** Title & Description, Visibility (`PUBLIC` vs `PRIVATE`), Type (`REGULAR` solo vs `INFLUENCER` group battle), GMT Start Date & Time
Votes, Prize Pool USDT, Registered Player Count, Gate Codes, Actions (`Edit`, `Apply Template`, `Delete`).

2. Create Tournament Modal & Form Field Contracts:

Form Field	Type & Constraints	Behavior & Database Target
Tournament Title	String (3–100 chars)	Display title on tournament lobby card & headers.
Description	Text (0–500 chars)	Rules, sponsor message, or narrative copy.
Visibility	PUBLIC PRIVATE	Private tournaments generate joinCode and require referral entry.
Tournament Type	REGULAR INFLUENCER	Influencer battles create 2–6 TournamentTeam rows.
Group / Team Count	Integer (2 to 6)	Number of competing influencer-led teams.
GMT Start Date	Date (UTC Midnight)	Target calendar date for tournament match.
Time Slots Array	JSON Array (e.g. `[‘14:00’, ‘18:00’, ‘21:00’]`)	Candidate GMT time options for player voting.
Prize Pool USDT	String / Base Units	Total guaranteed prize pool (e.g. ‘5,000 USDT’).
Winner Count	Integer (1, 3, or N)	Number of top finishers eligible for prize payout.
Player Capacity (Min/Max)	Integers (e.g. Min 2, Max 100)	Triggers automated AI bot backfill if min not reached.

3. Campaign & Cause Templates Gallery:

Showcases all 7 standard presets (Nike 31, Adidas 31, MooMorrow, Cyber Neon, Oasis Water, Obsidian Wildlife, Candy Smiles) plus custom saved templates. Each card displays visual color swatches, charity metrics, and a **[Apply to Tournament]** 1-click deployer button.

4. Tab 3: Sponsor Campaign & Cause Studio Deep Dive
(`/admin/tournament/[id]/campaign`)

The enterprise co-branding studio implements 5 interconnected visual and functional customization panels:

Studio Customization Panel	Configurable Parameters & Visual Controls	Live Frontend Rendering Target
1. Identity & Copy	• Campaign Title: Main promotional event name. • Sponsor Name: Entity name (e.g. Nike, Adidas, MooMorrow). • Disclosure Label: ‘Powered by’ ‘Presented by’ ‘Sponsored by’. • Feature Panel Copy: Headline & body description.	Rendered in header subtitle pill, sidebar feature box, and tournament lobby card.
2. Visual Theme & Colors	• Primary Color: Header typography & card text. • Secondary Color: Accent borders, glow shadows, badges. • Desktop Arena Backdrop (16:9): Upload / preset selector. • Mobile Backdrop (9:16): Portrait mobile arena background. • Overlay Opacity: Slider from 0.0 to 1.0 (dark tint).	Applies directly to desktop 3-column arena background and mobile arena shell.
3. Logo Tile & 3D Motion	• Logo Text: Fallback typography when media is absent. • Media Upload: PNG, SVG, or MP4 video. • Motion Presets: `pulse` (breathing scale), `turntable` (3D horizontal spin), `float` (vertical bobbing), `static`. • Desktop / Mobile Toggles: Independent slot toggles.	Rendered inside the sponsor-themed card in the header marquee.
4. Charity Cause & Giving	• Enable Toggle: Activates charity tracking layer. • Cause Title & Label: Category badge & project name. • Beneficiary Trust: Registered non-profit organization. • Fundraising Goal & Current Raised: Target amount & tally. • Currency: USD, USDT, EUR, GBP, CAD. • CTA Button & Outbound URL: Verified external link.	Rendered in single Heart Cause header button (`HeartHandshake`), expanding cause drawer, and lobby progress bar.
5. Versioning & Rollback	• Working Draft: Auto-saved revisions with revision counter. • Review Pipeline: Multi-lane `BRAND` & `SAFETY` approvals. • 1-Click Rollback: Clones historical version into new published revision. • Emergency Kill Switches: `isPaused` & `isCausePaused`.	Controls real-time live propagation and safety rollbacks.

5. Tab 4: Sponsor Accounts & Partner Management (`/admin/sponsor`)

Enables administrators to create and manage dedicated, isolated partner sponsor accounts:

- **Onboarding Console:** Admin enters Sponsor Name, optional custom Username, and optional initial Password (or auto-generates a secure 12-char password).
- **Secret-Box Encryption at Rest:** In the Sponsor table, passwordHash is hashed with Argon2id for login verification, while passwordEnc stores a

- reversibly-encrypted copy encrypted using AES-256-GCM (secret-box). This allows the admin to reveal or copy credentials when onboarding clients.
- **Sponsor Management Table:** Displays all registered sponsors, assigned tournaments count, status toggle (Active vs Banned), password reveal toggle (eye icon), and 1-click clipboard copy.
 - **Sponsor Portal Isolation:** Sponsors sign in at /sponsor/login and receive a scoped JWT restricted exclusively to their assigned tournaments.

6. Tab 5: Game Studio & Platform Rules Engine (`/admin/game`)

The Game Studio (AdminGameStudio.tsx) manages global game configuration across 7 sub-tabs:

Sub-Tab	Configuration Settings & Parameters	Real-World Impact on Match Engine
1. Control Room	Live interactive arena simulation preview.	Real-time visual testbed for game rules and cosmetics.
2. Brand & Copy	Game Title, Subtitle, Marquee Announcement, Victory Quote.	Updates `branding.gameTitle` and announcement text across all clients.
3. Arena Designer	World backdrop selector, primary/secondary colors, lighting, particles.	Configures global default arena environment.
4. Game Setup	Classic mode toggle, Skills mode toggle, Turn seconds (3–30s), Default bot count (1–100), Guest name prompt, Max name length.	Enforces turn timer duration and player validation rules in `CountdownGateway`.
5. Skill Deck	Drag-to-reorder skills, add/remove skills, edit skill name, icon, cooldown, effect multiplier, description.	Controls available skill powers in 2-skill pre-match loadout selector.
6. Bot Roster	Bot names, avatar variants, skill loadouts, AI difficulty weighting.	Defines names and behaviors of AI opponents when backfilling matches.
7. Player Menu	Navigation builder: links, icons, protected auth requirements.	Customizes mobile bottom navigation and sidebar items.

7. Tabs 6 & 7: User Moderation, Ledger Auditing & Global Settings

- **User Moderation (`/admin/users`):** Search players by email, username, or Solana address. Actions: Ban/Unban account (flips User.status and increments tokenVersion to revoke all active sessions immediately), trigger password reset email.
- **Ledger Adjustment & Auditing:** Admin can adjust a user's USDT wallet balance. Every adjustment writes an append-only row to LedgerEntry with type='ADJUSTMENT' and refType='ADMIN_AUDIT:{adminId}' to maintain complete financial reconciliation.
- **Theme Family Switcher (`/admin/settings`):** Switches platform visual world: **Barnaby Pasture (Monster Mayhem)** (lush green pasture, gold finishes) vs **Moonlit Bazaar (Desert)** (amber night, desert sands). Broadcasts theme change over WebSockets.
- **Platform Maintenance Mode:** When enabled, blocks non-admin logins and displays maintenance screen across frontend.

8. Complete Admin Panel REST API Routing Catalogue

Domain	Endpoint & HTTP Method	Auth Guard & Detailed Description
Overview	GET /admin/promo-tournaments/overview GET /admin/inquiries	AdminGuard. Aggregates live player stats, active sponsors count, pending approvals, and partner inquiries.
Tournaments	GET /admin/promo-tournaments POST /admin/promo-tournaments PATCH /admin/promo-tournaments/:id DELETE /admin/promo-tournaments/:id POST /admin/.../:id/status	AdminGuard. Full tournament CRUD, gate codes generator, status approvals, cascading record deletion.
Campaign Studio	GET /admin/.../campaign PATCH /admin/.../draft POST /admin/.../assets/:kind POST /admin/.../publish POST /admin/.../rollback	AdminOrSponsor. Working drafts, media uploads, BRAND/SAFETY review records, 1-click rollback, emergency kill switches.
Templates	GET /admin/campaign-templates POST /admin/campaign-templates POST /admin/.../apply-template	AdminGuard. Global presets (Nike 31, Adidas 31, etc.) and atomic 1-click tournament deployment.
Sponsors	GET /admin/sponsors POST /admin/sponsors PATCH /admin/sponsors/:id	AdminGuard. Provision sponsor accounts with secret-box encrypted credentials and tournament ownership linking.
Game Config	GET /admin/game-config POST /admin/game-config POST /admin/game-config/reset	AdminGuard. 7-tab game engine rules, turn seconds, bot roster, skill deck, and branding copy persistence.
Users & Ledger	GET /admin/users POST /admin/users/:id/ban POST /admin/users/:id/adjust	AdminGuard. User search/filter, ban/unban moderation, token revocation, wallet ledger adjustments.